Pipe Bend Thickness Report

ASME B31.3-2018 para. 304.2.1 | Pipe bend under internal pressure

DESIGN EQUATION

ASME B31.3-2018, 304.2.1, (3c)-(3e)
t = P D / [2 (S E W / I + P Y)]; tm = t + c
I_in = [4(R1/D) - 1] / [4(R1/D) - 2]
I_out = [4(R1/D) + 1] / [4(R1/D) + 2]
Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

Project	**QA — مثال كوع**
Prepared by	**Device preview**
Design reference	**B31.3-2018 QA**
Report date	**2026-10-21 16:36**

01 BEND GEOMETRY

Material / grade	**User coefficients - example only**
Outside diameter D	**114.3 mm**
Bend radius to centreline R1	**171.45 mm (1.5 D)**
Inspection / pipe reference	**QA-UT**

02 DESIGN INPUTS

Internal design gauge pressure P	**20 bar(g) = 2 MPa**
Allowable stress S	**100 MPa**
Design metal temperature	**20 C**
Stress source	**B31.3-2018 A-1/A-1M; S supplied at design temperature**
Quality factor E / weld factor W	**0.85 / 0.9**
Coefficient Y	**0.4**
Corrosion / erosion allowance	**1.5 / 0.5 mm**
Mechanical removal allowance	**1 mm (before fabrication)**

03 CALCULATION TRACE

Use the equations above with the following geometry factors. The intrados factor is above 1 and governs the pressure wall.

Intrados factor I	**1.25**
Extrados factor I	**0.875**
Pressure thickness at intrados	**1.8435 mm**
Pressure thickness at extrados	**1.2955 mm**
Matching straight pipe, I = 1	**1.4787 mm**
Bend wall before fabrication	**4.8435 mm**
Required remaining wall at intrados	**3.8435 mm**

04 MEASURED INTRADOS WALL

Minimum measured intrados wall	**4 mm**
Required remaining wall	**3.8435 mm**
Measured minus required	**0.1565 mm**

MEETS the internal-pressure wall requirement for these inputs.

05 LIMITS AND REFERENCES

Basis: ASME B31.3-2018, 304.2.1. Verify user-supplied S, E, W, Y and allowances against the code and project specification. This report checks required bend wall; it does not predict forming or geometric thinning. Excluded: miter bends (304.2.3), external pressure, other sustained/occasional loads, fatigue and remaining-life assessment.